BRAYDEN HUGUENARD

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY | GPA: 4.0/4.0 *Atlanta, Georgia*

M.S. Computer Science | Machine Learning *Graduation: May 2028*

B.S. Computer Science | Intelligence & Modeling and Simulation *Graduation: December 2026*

- Relevant Courses: ML, Intro to AI, Computer Vision, Deep Learning, Databases, Data Structures & Algorithms, Linear Algebra
- Recipient of Zell Miller (2024, 2025, 2026), Daniel Hager (2024), and Delta Air Lines Academic (2025, 2026) Scholarships

SKILLS

Programming Languages: Python, Swift, SQL, Java, C, C++, JavaScript, HTML

Frameworks & Libraries: SwiftUI, React, FastAPI, PyTorch, scikit-learn, LangChain, FAISS, Pandas, NumPy, SwiftData, CoreML

Cloud, Tools & Platforms: AWS (Lambda, Bedrock, S3, ECS, CloudWatch), MongoDB, Terraform, Docker, Firebase, Supabase

EXPERIENCE

iOS Engineer (Contract) | *auris, Midtown ENT* *September 2026 – Present*

- Designed and built auris end to end as the sole engineer, an iOS app for ENT patients to track tinnitus severity and daily habits
- Developed trend analysis to surface associations between habits and symptoms with CSV/PDF exports for clinic visits
- Built an opt-in AI summary/chat API on AWS (Lambda, Bedrock, DynamoDB) with Supabase JWT auth and response caching
- Implemented local-first privacy controls: on-device health data, no server chat retention, and account deletion in Supabase/AWS

Technical Lead | *Georgia Tech App* *December 2025 – Present*

- Architected SwiftUI codebase and established custom SwiftLint standards, aligning a 25+ person team on a unified approach
- Integrated 6 GT services across dining, transit, capacity, and events through 13+ API endpoints and ingestion pipelines
- Conducted user interviews to uncover usability issues and translate user feedback into product design and feature decisions
- Built adaptive polling for live bus tracking, reducing unnecessary network requests while maintaining real-time ETA updates
- Integrated GT SSO (CAS/OIDC) for secure backend access and built API pipelines for live dining, capacity, and event data
- Led collaboration across SGA, OIT, Dining, Library, and Institute Communications to integrate campus services into the app
- Partnered with SGA as the app helped shape a new committee supporting student-built technology projects across campus

Teaching Assistant | *CS4400 – Intro to Database Systems* *August 2026 – Present*

- Develop and administer database coding exams for 500+ students and host weekly office hours and technical review sessions
- Evaluate semester-long database projects through required demos, providing feedback on design and implementation

AI Innovation Intern | *Assent* *May 2026 – August 2026*

- Independently owned an internal LLM evaluation platform supporting 15 regulations, from prototype through deployment
- Built offline evaluation tool benchmarking model changes across 90+ labeled documents using accuracy and F1
- Improved evaluation accuracy from 64% to 81% through iterative criteria and methodology refinement with SMEs
- Reduced per-document token usage ~40%, from ~19K to ~11K, by consolidating common criteria and skipping redundant checks
- Evaluated Lambda and FastAPI architectures, deploying FastAPI on ECS with Terraform to support differing team requirements
- Built a 248-test hermetic suite covering end-to-end evaluation, API, and data-processing workflows with a 100% pass rate
- Connected three engineering teams with compliance SMEs, driving collaboration and adoption of the evaluation platform

Teaching Assistant | *CS3600 – Intro to Artificial Intelligence* *August 2025 – May 2026*

- Led 3 TAs developing review materials for 600+ students across two semesters, including practice resources and course reviews
- Held weekly office hours teaching AI concepts while reviewing and improving homework assignments, exams, and solutions

Software Engineer | *PlayOn Sports (Client Capstone)* *January 2026 – April 2026*

- Developed a Tree-of-Thought text-to-SQL pipeline generating 3 parallel queries and using an LLM judge to select the best
- Built SQL safety validation blocking 9 write/DDL operations before execution, enforcing read-only queries with a 200-row cap
- Engineered dynamic schema inference and chunked concurrent ingestion for CSV datasets exceeding 600K rows
- Built a custom SVG chart-rendering engine with axis scaling and chart geometry, enabling one-click PDF report generation
- Placed 2nd overall among 40+ teams at the GT Capstone Expo for technical execution and project presentation
- Improved text-to-SQL accuracy by 23 points, from 68% to 91%, through targeted prompt engineering for complex query handling

Technical Lead | *Georgia Tech iOS Club* *July 2025 – November 2025*

- Led and mentored 25+ developers building Reverie, winning “Best App” and “Best Design” at Demo Day
- Built an on-device AI pipeline using Apple Foundation Models and Stable Diffusion/CoreML to generate dream imagery
- Accelerated on-device image generation ~6×, from 30+ seconds to ~5 seconds, through model caching and concurrency
- Built a Vision-based image pipeline to create stickers from generated dreams, with persistent image storage through Firebase

Developer | *Georgia Tech iOS Club* *August 2024 – December 2024*

- Built SwiftUI stop-finding and itinerary workflows for Nomad, integrating Yelp, Codable models, search filters, and geocoding
- Contributed 55+ commits to a 20+ developer codebase, winning “Best App” and “Most Popular” at iOS Club Demo Day

PROJECTS

Teen Mental Health Predictive Model *September 2025 – December 2025*

- Diagnosed a synthetic dataset with no learnable signal and rebuilt the ML pipeline on validated teen depression data
- Built and tuned Random Forest and SVM models achieving ~85% accuracy and ~0.88 F1 with cross-validation
- Identified 2 dominant Random Forest features that preserved model performance while reducing training time by ~80%